OVER A JACOBSON BASE THE FINITE TYPE POINTS HAVE TRANSCENDENCE DEGREE ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a presentation reduction that produces a closed point in the presentation and invokes the scheme case; the shrinking step is justified by the cited identification of finite type points.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Lifting finite type points along a locally finite type surjective morphism. Lemma lemma-finite-type-points-surjective-morphism (44058), used at proof lines 6824-6825. Produces the finite type point u of U over x.
- Finite type points of a scheme can be made closed after shrinking. Morphisms, Lemma morphisms-lemma-identify-finite-type-points, used at proof lines 6826-6827. Makes u a closed point of the shrunk U.
- The scheme case over a Jacobson base. Morphisms, Lemma morphisms-lemma-jacobson-finite-type-points, used at proof lines 6828-6830. Gives that kappa(u)/kappa(v) is finite.
- The definition of Jacobson for an algebraic space. Properties of Spaces, Remark spaces-properties-remark-list-properties-local-etale-topology, used at proof lines 6814-6816. Fixes the hypothesis on Y.

Source reference. Original lemma 06EJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8218–8218. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02J2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3321–3321. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01TB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2368–2368. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark 03E7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4446–4446. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is locally of finite type, Y is Jacobson (Properties of Spaces, Remark 03E7), and $x \in |X|$ is a finite type point of X, then the transcendence degree of $x/f(x)$ is 0.*

Proof. Choose a scheme V and a surjective étale morphism $V \to Y$. Choose a scheme U and a surjective étale morphism $U \to X \times_Y V$. By Lemma 06EJ we can find a finite type point $u \in U$ mapping to x. After shrinking U we may assume $u \in U$ is closed (Morphisms, Lemma 02J2). Let $v \in V$ be the image of u. By Morphisms, Lemma 01TB the extension $\kappa(u)/\kappa(v)$ is finite. This finishes the proof. □